Lecture 23

# Antiderivatives and integration rules

**Uses.**

Summary 2: Real powers and logarithms have their stated domains. An interval contains every point between any two of its points.
Summary 7: A derivative is a difference-quotient limit. Differentiable functions are continuous.
Summary 8: The sum, product and chain rules hold. The derivatives of real powers, logarithms, exponentials, trigonometric functions and their stated inverse branches are known.
Summary 9: A function with zero derivative throughout an interval is constant, by the mean value theorem.

**Objectives.** *An antiderivative reverses differentiation on a specified interval. Its ambiguity is one additive constant on that interval. The chain rule gives substitution, and the product rule gives integration by parts. Every formula we obtain can be verified by differentiating it on the original domain.*

By the end of this lecture you will be able to:

- Find the complete antiderivative family on an interval and distinguish constants on separate domain intervals.
- Apply substitution with the inner derivative and its domain restrictions retained.
- Prove and use the antiderivative form of integration by parts.
- Select a primitive from one prescribed value and verify the resulting function by differentiation.

## 23.1 A primitive and its additive constant

Let $I \subseteq \mathbb{R}$ be a nonempty open interval and let $f : I \to \mathbb{R}$. An *antiderivative* (also called a primitive) of $f$ on $I$ is a differentiable function $F : I \to \mathbb{R}$ satisfying

$$F'(x) = f(x) \qquad \text{for every } x \in I.$$

On a union of disjoint open intervals, a primitive is a function differentiable at every point of the union with derivative $f$ there. The interval theorem below applies to each interval separately. For example, $F(x) = x^3/3$ is a primitive of $f(x) = x^2$ on $\mathbb{R}$ because $F'(x) = x^2$ at every real point. Adding any constant $C \in \mathbb{R}$ gives another primitive, since the derivative of a constant is zero.

**Theorem 23.1** (All primitives on an interval)**.** *If $F$ is a primitive of $f$ on $I$, every primitive of $f$ on that interval is $F + C$ for exactly one constant $C \in \mathbb{R}$.*

*Proof.* Let $G$ be any other primitive. The difference $G - F$ has derivative $G' - F' = f - f = 0$ on $I$. For any two points $a < b$ in $I$, their closed interval lies in $I$. The difference is continuous on $[a, b]$ and differentiable on $(a, b)$. The mean value theorem from Lecture 9 makes its values at $a$ and $b$ equal, since every derivative value is zero. Thus $G - F$ is one constant on all of $I$. Conversely, differentiating $F + C$ gives $f$, so every constant produces a primitive. If two constants gave the same function, evaluating at any point of the nonempty interval would make the constants equal. ∎

The *indefinite integral* denotes this whole primitive family. We write it in the form

$$\int f(x)\,\mathrm{d}x = F(x) + C, \qquad C \in \mathbb{R}, \tag{23.1}$$

with the interval supplied by the statement. This notation is introduced through differentiation. It does not yet assign an area or a numerical value to the symbol. The expression $\mathrm{d}x$ records the variable with respect to which the primitive is differentiated. The constant belongs to the whole answer, rather than to the integrand.

The interval restriction matters. On either half-line $x > 0$ or $x < 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}.$$

For $x > 0$ this is the logarithm derivative of Lecture 8. For $x < 0$, write $\ln|x| = \ln(-x)$ and apply the chain rule: its derivative is $(1/(-x))(-1) = 1/x$. Thus the primitives of $1/x$ on $\mathbb{R} \setminus \{0\}$ have the form

$$F(x) = \begin{cases} \ln(-x) + C_-, & x < 0, \\ \ln x + C_+, & x > 0, \end{cases} \qquad C_-, C_+ \in \mathbb{R}.$$

There is no derivative condition connecting the two half-lines, since zero is outside the domain. The uniqueness theorem applies separately to each interval. It does not force $C_- = C_+$.

A prescribed value selects one member of a primitive family on an interval. If $x_0 \in I$ and $F(x_0) = y_0$ is required, start with one primitive $F_0$. Every candidate is $F_0 + C$, and the value forces $C = y_0 - F_0(x_0)$. Hence the unique function meeting both requirements is

$$F(x) = F_0(x) + y_0 - F_0(x_0).$$

Differentiation gives the required derivative, and evaluation at $x_0$ gives the prescribed value. On a disconnected domain, a value in one interval fixes only that interval's constant; constants on other intervals remain independent.

## 23.2 A table and linear combinations

The derivative formulas of Lecture 8 immediately give the following primitives. The third column states where the displayed derivative is valid. Any interval used must lie entirely in that domain.

| $f(x)$ | one primitive $F(x)$ | domain restrictions |
|---|---|---|
| $x^\alpha$ | $x^{\alpha+1}/(\alpha+1)$ | $x>0, \quad \alpha \in \mathbb{R}\setminus\{-1\}$ |
| $1/x$ | $\ln\lvert x\rvert$ | $x \neq 0$ |
| $e^x$ | $e^x$ | $x \in \mathbb{R}$ |
| $\sin x$ | $-\cos x$ | $x \in \mathbb{R}$ |
| $\cos x$ | $\sin x$ | $x \in \mathbb{R}$ |
| $1/\cos^2 x$ | $\tan x$ | $\cos x \neq 0$ |
| $1/(1+x^2)$ | $\arctan x$ | $x \in \mathbb{R}$ |
| $1/\sqrt{1-x^2}$ | $\arcsin x$ | $-1<x<1$ |

Each answer is $F+C$, not just the representative in the middle column. For a nonnegative integer exponent $\alpha = n$, the power formula also holds on all of $\mathbb{R}$, by the polynomial derivative rule. The case $\alpha = -1$ belongs to the logarithm row instead; dividing by $\alpha+1$ would be undefined. The square root in the final integrand is positive on its stated open interval. Its endpoints are excluded because the integrand is undefined there. For the tangent row, work on one interval with no zero of the cosine. Crossing a missing point would incorrectly identify separate integration constants.

For example, the coefficient in the primitive of $e^{kx}$ is essential. If $k \in \mathbb{R}\setminus\{0\}$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{e^{kx}}{k}\right) = e^{kx}.$$

At $k=0$ the integrand is 1 and its primitives are $x+C$. The quotient formula cannot be used at that parameter. This distinction comes from the derivative, rather than from extending a formula through an undefined denominator.

If $F' = f$ and $G' = g$ on the same interval $I$, the sum and scalar rules give

$$(aF+bG)' = af+bg \qquad (a,b\in\mathbb{R}).$$

Therefore the complete family for $af+bg$ is

$$\int (af(x)+bg(x))\,\mathrm{d}x = aF(x)+bG(x)+C.$$

Completeness uses Theorem 23.1 after finding this one representative. The final constant is independent of $a$ and $b$. For instance, when $a=b=0$, all primitives are still the constant functions; they are not restricted to the zero function. When several primitive calculations are combined, their constant terms can be collected into this one arbitrary constant on $I$.

## 23.3 Substitution from the chain rule

**Theorem 23.2** (Antiderivative substitution). *Let $I$ and $J$ be nonempty open real intervals. Suppose $F: J\to\mathbb{R}$ is a primitive of $f: J\to\mathbb{R}$ and $g: I\to J$ is differentiable. Then all primitives on $I$ of $f(g(x))g'(x)$ are*

$$F(g(x))+C, \qquad C\in\mathbb{R}.$$

*Proof.* The range condition makes the composition defined throughout $I$. The chain rule gives $(F \circ g)'(x) = F'(g(x))g'(x) = f(g(x))g'(x)$ for every $x \in I$. Thus $F \circ g$ is one primitive of the stated integrand. The interval uniqueness theorem gives exactly the added-constant family. ∎

In substitution notation, choose $u = g(x)$ and write $\mathrm{d}u = g'(x)\mathrm{d}x$. This records the chain-rule factor in the theorem. It does not permit division by a derivative that may be zero. The theorem requires neither monotonicity nor a nonzero derivative of $g$. The composition must be defined, and the complete inner-derivative factor must remain in the integrand. After finding a primitive in the variable $u$, substitute back to obtain a function on the original interval in $x$.

**Example 23.3** (A radical and a prescribed value)**.** Find all primitives on $\mathbb{R}$ of

$$f(x) = (6x - 4)\sqrt{3x^2 - 4x + 5},$$

and select the one with value 2 at $x = 1$. Set $u = g(x) = 3x^2 - 4x + 5$, whose derivative is $6x - 4$. Completing the square gives

$$g(x) = 3\left(x - \frac{2}{3}\right)^2 + \frac{11}{3} > 0 \qquad (x \in \mathbb{R}).$$

The square-root primitive is therefore valid throughout the image of $g$. The substitution and power row give

$$\int (6x - 4)\sqrt{3x^2 - 4x + 5}\,\mathrm{d}x = \frac{2}{3}(3x^2 - 4x + 5)^{3/2} + C.$$

Here the derivative of the representative is $\frac{2}{3}\frac{3}{2}g(x)^{1/2}(6x - 4) = f(x)$, including at $x = 2/3$, where the inner derivative vanishes. At $x = 1$, the inner value is 4, so the required function must have $16/3 + C = 2$. Hence the selected primitive is

$$F(x) = \frac{2}{3}(3x^2 - 4x + 5)^{3/2} - \frac{10}{3}.$$

It has the prescribed value and is unique on $\mathbb{R}$ by the interval theorem.

The rational integrand $x/(x^2 - 4)$ has domain $\mathbb{R} \setminus \{-2, 2\}$. Write its domain as the three intervals $I_1 = (-\infty, -2)$, $I_2 = (-2, 2)$ and $I_3 = (2, \infty)$. Figure 23.1 separates the constants belonging to those intervals.

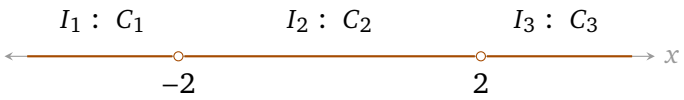



Figure 23.1: The missing endpoints divide the domain into three intervals. A derivative condition leaves one independent additive constant on each interval.

**Example 23.4** (A logarithm on three intervals)**.** Find every primitive of $x/(x^2 - 4)$ on its domain. Take $u = x^2 - 4$, so the required derivative factor is $2x$ and the integrand is one half of $g'(x)/g(x)$. On $I_1$ and $I_3$ the value $u$ is positive; on $I_2$ it is negative. The logarithm row supplies a primitive on either appropriate half-line. Substitution gives the representative

$$F_0(x) = \frac{1}{2}\ln|x^2 - 4|.$$

Its derivative is $\frac{1}{2}(2x)/(x^2-4) = x/(x^2-4)$ at every domain point. In particular, the formula is valid at $x = 0$; no step has divided by $x$. The full family is

$$F(x) = F_0(x) + C_j \quad \text{for } x \in I_j, \qquad j = 1, 2, 3,$$

where $C_1, C_2, C_3 \in \mathbb{R}$ are independent. Each domain interval forces one constant by Theorem 23.1, and arbitrary choices of those constants give differentiable functions on the whole stated domain. No continuity at the excluded endpoints is required.

**Example 23.5** (A missing numerical factor)**.** Compute the primitive family on $\mathbb{R}$ for

$$\frac{e^{2x}}{1+e^{4x}}.$$

Choose $u = e^{2x}$, so $\mathrm{d}u = 2e^{2x}\mathrm{d}x$. The outer primitive is $\arctan u$, but the original numerator contains only half of the required derivative factor. Consequently

$$\int \frac{e^{2x}}{1+e^{4x}}\,\mathrm{d}x = \frac{1}{2}\arctan(e^{2x}) + C.$$

Differentiation gives $\frac{1}{2}(1/(1+e^{4x}))2e^{2x}$, which is the original integrand. The positive exponential has image in $(0,\infty)$, contained in the real domain of the arctangent primitive. The formula is valid for every $x \in \mathbb{R}$.

For a trigonometric example, set $u = \cos x$ in $\int \sin x \cos^3 x\,\mathrm{d}x$. The derivative factor is $-\sin x$, so the primitive family is

$$\int \sin x \cos^3 x\,\mathrm{d}x = -\frac{\cos^4 x}{4} + C \qquad (x \in \mathbb{R}).$$

Its derivative is $\sin x \cos^3 x$ by the chain rule. The cosine is not one-to-one on $\mathbb{R}$ and its derivative has many zeros. Neither prevents the primitive formula, because the outer polynomial primitive is defined on all of $\mathbb{R}$.

### A proposed substitution must survive differentiation

On $(0,\infty)$, consider the proposed primitive $H(x) = e^{x^2}/(2x)$ for $e^{x^2}$. Dividing the exponential by the inner derivative is not a substitution rule. The quotient and chain rules give

$$H'(x) = e^{x^2} - \frac{e^{x^2}}{2x^2}.$$

The second term is nonzero on the stated interval, so the candidate fails. An added constant cannot correct a wrong derivative. On that positive interval, $u = x^2$ has inverse branch $x = \sqrt{u}$. The inverse derivative gives the transformed factor $\mathrm{d}x = \mathrm{d}u/(2\sqrt{u})$, not a constant multiple of $\mathrm{d}u$. The transformed integrand contains $e^u/(2\sqrt{u})$, rather than just $e^u$. The displayed table does not supply a primitive for that expression. This check rejects the proposed formula without claiming that the integral has no primitive. Existence for continuous integrands is proved in Lecture 26.

## 23.4 Integration by parts from the product rule

**Theorem 23.6** (Antiderivative integration by parts). *Let $u, v : I \to \mathbb{R}$ be differentiable on a nonempty open interval. If $H$ is a primitive of $u'v$ on $I$, all primitives of $uv'$ on $I$ are*

$$u(x)v(x) - H(x) + C, \qquad C \in \mathbb{R}.$$

*Conversely, a primitive of $uv'$ produces one of $u'v$ by subtracting it from $uv$. Thus one of those two integrands has a primitive exactly when the other does.*

*Proof.* The product rule gives $(uv)' = u'v + uv'$. If $H' = u'v$, subtraction gives $(uv - H)' = uv'$. This proves the proposed representative, and the interval uniqueness theorem gives the complete added-constant family. Conversely, if $G' = uv'$, the same product rule gives $(uv - G)' = u'v$. This proves the converse existence statement. No continuity of the derivatives was used; the differentiability and stated primitive hypotheses supply all required steps. ∎

To apply the theorem, choose a factor $u$ and a remaining factor $v'$. Find a primitive $v$ of that remaining factor. Differentiate $u$, multiply it by $v$ and find a primitive $H$ of $u'v$. The answer is then $uv - H + C$. The product term and the minus sign follow from the product rule, so they must be retained even when the second primitive is simple. This rule is called *integration by parts*. It replaces one primitive problem by another; the choice is useful when the new integrand is easier to handle.

**Example 23.7** (A polynomial factor and a cosine). Find the primitive family on $\mathbb{R}$ for $x\cos(2x)$. Choose $u(x) = x$ and $v'(x) = \cos(2x)$. Then $u'(x) = 1$ and $v(x) = \sin(2x)/2$ by the chain-rule factor. One primitive of the new product $u'v = \sin(2x)/2$ is $H(x) = -\cos(2x)/4$. The theorem therefore gives

$$\int x\cos(2x)\,\mathrm{d}x = \frac{x\sin(2x)}{2} + \frac{\cos(2x)}{4} + C.$$

For a verification, the derivative of the first term is $\sin(2x)/2 + x\cos(2x)$ and that of the second is $-\sin(2x)/2$. The sine terms cancel, leaving the required $x\cos(2x)$ at every real point. Both numerical factors came from differentiating the argument $2x$. There is one arbitrary constant because the interval is all of $\mathbb{R}$.

**Example 23.8** (Combining parts with substitution). Find the complete primitive family of $\arctan x$ on $\mathbb{R}$. Write the integrand as $(\arctan x)\cdot 1$ to choose factors for parts. Take $u(x) = \arctan x$ and $v'(x) = 1$. Their derivative and primitive are $u'(x) = 1/(1+x^2)$ and $v(x) = x$. The new integrand to be integrated is therefore $u'(x)v(x) = x/(1+x^2)$.

This second primitive follows by substitution rather than another application of parts. Set $w = 1 + x^2$, so $\mathrm{d}w = 2x\mathrm{d}x$. The image is positive for every real $x$, and the logarithm primitive gives

$$H(x) = \frac{1}{2}\ln(1+x^2), \qquad H'(x) = \frac{x}{1+x^2}.$$

There is no absolute-value branch to choose because $1 + x^2 > 0$ throughout the original interval. The parts theorem now gives the complete answer

$$\int \arctan x\,\mathrm{d}x = x\arctan x - \frac{1}{2}\ln(1+x^2) + C, \qquad x \in \mathbb{R}.$$

To verify it, the product term differentiates to $\arctan x + x/(1+x^2)$, while the logarithmic term differentiates to $-x/(1+x^2)$. Their sum is exactly the required integrand. The construction

uses both rules with their full derivative factors: parts creates the rational integrand, and substitution supplies its logarithm. The interval theorem then gives every primitive through the one constant $C$.

Sheet 23 proves a sine-power reduction formula using this same product rule and replacement of $\cos^2 x$ by $1 - \sin^2 x$. Lecture 24 develops repeated and cyclic parts calculations and rational integration. Motion recovered from acceleration is treated in Lecture 32. These later applications use the primitive rules proved here.

**Summary 23.**

- Theorem 23.1: once one primitive is known on a nonempty open interval, all are $F + C$. Separate domain intervals have independent constants. A value at one point fixes that interval's constant.
- § 23.2: the table reverses the known derivative formulas on their stated domains. Real-power, logarithm and trigonometric restrictions remain part of each answer.
- Theorem 23.2: for differentiable $g : I \to J$ and a primitive $F$ of $f$ on $J$, the primitive family of $f(g)g'$ on $I$ is $F \circ g + C$. Monotonicity and a nonzero inner derivative are not required.
- Theorem 23.6: if $H' = u'v$ for differentiable $u$ and $v$, the primitives of $uv'$ are $uv - H + C$ on the same interval. Every proposed answer can be tested by differentiating on its original domain.

Sheet 23 tests substitution, parts, constants and the sine-power recurrence. Lecture 24 combines these rules into further integration techniques.

## References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§4.9,5.5 and 7.1. V. A. Zorich, *Mathematical Analysis I*, 2nd ed., Springer, 2015, §5.7.